Indian Institute of Technology, Kanpur

Department of SPASE

Laboratory Report

Radio Instrumentation: RF Transmission, Reflections, and Standing Waves

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1 Objective

(a) To determine the phase velocity and velocity factor of a signal in a coaxial cable from the frequency spacing between consecutive maxima or minima produced by reflection from an open-ended cable.

(b) To study signal reflection from an open-ended cable and from a cable terminated with a 50 Ω load.

(c) To estimate the magnitude of the reflection coefficient and verify that the coaxial cable is approximately matched to a 50 Ω termination, indicating that its characteristic impedance is close to 50 Ω.

(d) To calculate the inductance and capacitance per unit length using the measured phase velocity and the approximate value $Z_0 \approx 50\ \Omega$.

(e) To measure the insertion loss and attenuation per unit length of the coaxial cable.

2 Theory

Transmission lines are structures through which electromagnetic waves are propagated from one point to another. A coaxial cable is one such transmission line which propagates transverse electromagnetic (TEM) waves through it where both the electric and magnetic fields are transverse to the direction of propagation. The velocity of electromagnetic waves inside a transmission line is less than that of velocity of light in vacuum due to the presence of a dielectric medium between the conducting wire. Also, if there is a change in the medium, then the waves get reflected back, creating standing waves at times. The goal is to study these properties of a transmission line.

2.1 Phase velocity

If the physical length of coaxial cable under study is l and the phase velocity in the cable is v_p, then the reflected signal travels a round-trip distance $2l$. The time taken would be

$$\tau = \frac{2l}{v_p}$$

and the round-trip phase of the reflected signal would be

$$\phi_{RT} = 2\pi f\tau \ + \ \phi_\Gamma = \frac{4\pi f l}{v_p} \ + \ \phi_\Gamma$$

where ϕ_Γ is the phase shift introduced by the termination. For an ideal open circuit, $\phi_\Gamma = 0$. Two consecutive maxima, or two consecutive minima, are separated by a round-trip phase change of 2π. Thus,

$$2\pi\Delta f\tau = 2\pi \implies \tau = \frac{1}{\Delta f}$$

we get,

$$v_p = 2l\Delta f \tag{1}$$

The velocity factor of the cable is

$$V_F = \frac{v_p}{c}$$

where c $\approx 3 \times 10^8 m/s$ and Δf is the frequency difference between two consecutive maxima and minima.

2.2 Reflection coefficient and characteristic impedance

The voltage reflection coefficient at the load is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

where Z_L is the load impedance and Z_0 is the characteristic impedance of the cable. Special cases are such that

1. Open circuit: $Z_L \to \infty$ and $\Gamma = +1$.
2. Short circuit: $Z_L = 0$ and $\Gamma = -1$.
3. Matched load: $Z_L = Z_0$ and $\Gamma = 0$.

During a frequency sweep, the oscilloscope displays the superposition of the incident and reflected waves:

$$V_{osc} = V_{incident} + V_{reflected}$$

When the incident and reflected waves are in phase, they interfere constructively and the measured amplitude is maximum:

$$V_{max} = |V_{incident}| + |V_{reflected}|$$

When the incident and reflected waves are out of phase by 180°, they interfere destructively and the measured amplitude is minimum. Assuming the incident wave amplitude is greater than or equal to the reflected-wave amplitude,

$$V_{min} = |V_{incident}| - |V_{reflected}|$$

Adding and subtracting these two equations gives

$$|V_{incident}| = \frac{V_{max} + V_{min}}{2}; |V_{reflected}| = \frac{V_{max} - Vmin}{2}$$

Therefore, the magnitude of the reflection coefficient observed at the Tee junction can be estimated as

$$|\Gamma| = \frac{|V_{reflected}|}{|V_{incident}|} = \frac{V_{max} - V_{min}}{V_{max} + V_{min}} \tag{2}$$

2.3 Inductance and capacitance per unit length

Let L' denote the inductance per unit length in H/m and C' denote the capacitance per unit lenth in F/m. For a low-loss transmission line, $v_p = \frac{1}{\sqrt{L'C'}}$ and $Z_0 = \sqrt{\frac{L'}{C'}}$. Therefore,

$$L' = \frac{Z_0}{v_p} \tag{3}$$

and

$$C' = \frac{1}{Z_0 v_p} \tag{4}$$

2.4 Cable attenuation

Cable attenuation is the reduction in signal level as the wave propagates through the cable. In this experiment, power is measured in dBm using a spectrum analyzer. The insertion loss of the cable is

$$IL = P_{ref} - P_{cable} \tag{5}$$

where P_{ref} is the power measured using a short reference connection and P_{cable} is the power measured after inserting the cable under test. The attenuation per unit length is then

$$\alpha = \frac{IL}{l} dB/m \tag{6}$$

3 Components Required

RF signal generator, digital storage oscilloscope (DSO), spectrum analyzer, RF Tee connector, coaxial cable under test, short reference coaxial cable, 50Ω termination, open-circuit termination, suitable RF adapters and connectors.

4 Experimental Setup

Connect the components as shown in the figure below:

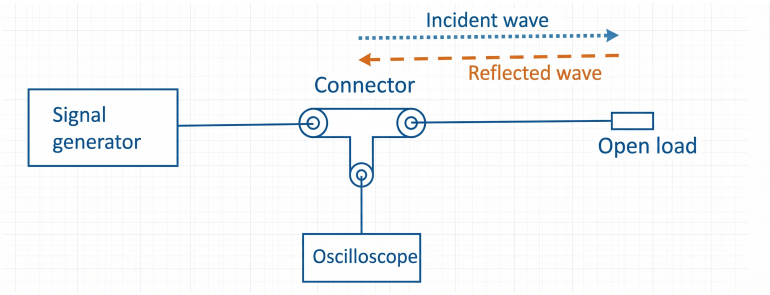


Figure 1: *Experimental Setup*

Certain precautions need to be taken before starting the experiment:

- Set the oscilloscope input impedance to 1 $M\Omega$. A 50 Ω oscilloscope input would load the Tee junction and alter the reflection pattern.
- Keep the signal-generator amplitude constant during each frequency sweep.
- Check whether the signal generator displays voltage for a 50 Ω load or for a high-impedance load. Use one setting consistently and record it.
- Use an input signal level that is clearly measurable but does not overdrive the oscilloscope or spectrum analyzer.

- Do not connect the signal generator directly to a short circuit without the cable and appropriate source protection.
- Use the same adapters and connector arrangement for reference and cable measurements wherever possible.

4.1 Phase velocity and velocity factor

1. Measure the physical cable length l.
2. Make connections as shown in Fig 1.
3. Leave the far end of the cable open.
4. Set the oscilloscope input impedance to 1 $M\Omega$.
5. Apply a sinusoidal signal of constant amplitude.
6. Sweep the frequency slowly over a range containing at least two maxima and two minima. Record the singla amplitude at suitable frequency intervals.
7. Plot the measured signal amplitude vs frequency and identify the frequencies f_1 and f_2 of two consecutive maxima or minima.
8. Calculate the frequency spacing: $\Delta f = |f_2 - f_1|$. Convert Δf to hertz before using it in the phase-velocity calculation.
9. Calculate the phase velocity and velocity factor as mentioned in Eq 1.

4.2 Reflection coefficient and approximate characteristic impedance

1. Retain the Tee-junction setup and use the same frequency range employed for the open-circuit measurement in section 4.1.
2. From the open-circuit data, determine $V_{max,open}$ and $V_{min,open}$ over one complete ripple period.
3. Calculate the magnitude of the reflection coefficient for the open-ended cable Γ_{open} using eq 2.
4. Connect the 50Ω termination to the far end of the cable.
5. Sweep over the same frequency range without changing the signal-generator amplitude.
6. Record $V_{max,50}$ and $V_{min,50}$.
7. Calculate $|\Gamma_{50}|$ using eq 2.
8. Compare Γ_{open} and Γ_{50}. If the reflection is substantially reduced with the 50Ω termination, conclude that the cable is approximately mathced to 50Ω and that its characteristic impedance is close to 50Ω.

4.3 Inductance and capacitance per unit length

Calculate the inductance and capacitance using eqs. 3 and 4 respectively.

4.4 Cable attenuation

1. Connect the signal generator to the spectrum analyzer using a short reference cable and the required adapters.
2. Set a suitable test frequency and input power. Keep both fixed throughout the measurement.
3. Record the reference power P_{ref} in dBm.
4. Replace the short reference connection with the cable under test while keeping the remaining setup unchanged.
5. Record the measured power P_{cable} in dBm.
6. Calculate the cable insertion loss using eq 5 and attenuation per unit length using eq 6.
7. Repeat at additional frequencies if the frequency depencence of attenuation is to be studied.

5 Observations

5.1 Phase velocity and velocity factor

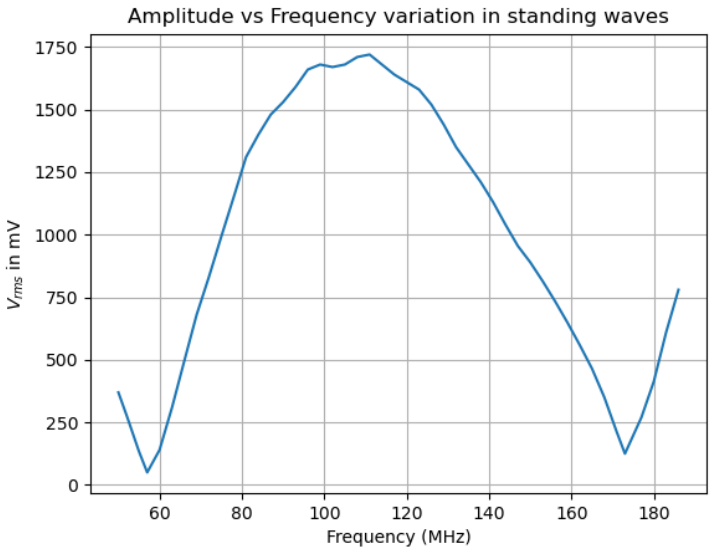


Figure 2: *Plot of amplitude versus frequency of standing waves.*

$$f_1 = 57 \text{ MHz}$$
$$f_2 = 173 \text{ MHz}$$
$$\implies \Delta f = 116 \text{ MHz}; \quad l = 92.5 \text{ cm}$$
$$\implies \text{Phase velocity, } v_p = 2.146 \times 10^8 \text{ m/s}$$
$$\text{Velocity factor, } V_F = 0.715$$

5.2 Reflection coefficient and characteristic impedance

$$\begin{aligned} V_{\text{max,open}} &= 1.72 \text{ V}; \quad V_{\text{min,open}} = 0.125 \text{ V} \\ V_{\text{max,50}} &= 0.928 \text{ V}; \quad V_{\text{min,50}} = 0.900 \text{ V} \\ \implies \text{Reflection coefficients, } \Gamma_{\text{open}} &= 0.864 \quad \text{and} \quad \Gamma_{50} = 0.015 \\ \text{and impedance, } Z_0 &= 48.5\ \Omega \end{aligned}$$

5.3 Inductance and capacitance per unit length

Now using Z_0 and v_p, we get

$$\begin{aligned} \text{Inductance, } L' &= 2.26 \times 10^{-7} H = 0.0226\ nH \\ \text{and Capacitance, } C' &= 9.61 \times 10^{-11} F = 96.1\ pF \end{aligned}$$

Thus, Inductance per unit length is $2.44 \times 10^{-7} H/m$ and capacitance per unit length is $1.04 \times 10^{-10} F/m$.

5.4 Cable attenuation

Frequency (MHz)	P_{ref}(dB)	P_{cable}(dB)	Insertion Loss (IL)	α
100	-0.57	-0.60	0.03	0.032
150	-0.79	-0.72	-0.07	-0.076
200	-0.91	-0.90	0.01	0.011

Table 1: *Cable attenuation at different frequencies.*

6 Error Analysis

The baseline reading uncertainties, based on instrumental least counts and typical experimental methodology, are taken as $\delta l = 0.1$ cm for the measuring tape and $\delta f = 1$ MHz for the individual frequency readings (yielding a combined spacing uncertainty of $\delta(\Delta f) = \sqrt{\delta f_1^2 + \delta f_2^2} = 1.41$ MHz). The voltage uncertainty is modeled as a 2% relative error ($\delta V = 0.02V$) because the oscilloscope scale was iteratively optimized during measurement. The 50 Ω termination is assumed to have a standard 1% physical manufacturing tolerance ($\delta Z_L = 0.5\ \Omega$).

6.1 Phase velocity

$$\frac{\delta v_p}{v_p} = \sqrt{\left(\frac{\delta l}{l}\right)^2 + \left(\frac{\delta(\Delta f)}{\Delta f}\right)^2} \tag{7}$$

Substituting the measured values $l = 0.925$ m ($\delta l = 0.001$ m) and $\Delta f = 116$ MHz:

$$\delta v_p = 2.146 \times 10^8 \times \sqrt{\left(\frac{0.001 \text{ m}}{0.925 \text{ m}}\right)^2 + \left(\frac{1.41 \text{ MHz}}{116 \text{ MHz}}\right)^2} \approx 0.026 \times 10^8 \text{ m/s}$$

Thus, the calculated phase velocity is $v_p = (2.15 \pm 0.03) \times 10^8$ m/s.

6.2 Reflection coefficient and impedance

$$\delta\Gamma = \frac{2}{(V_{max}+V_{min})^2}\sqrt{(V_{min}\delta V_{max})^2 + (V_{max}\delta V_{min})^2} = \frac{2\sqrt{2}\times 0.02\times V_{max}V_{min}}{(V_{max}+V_{min})^2} \tag{8}$$

For the open circuit ($V_{max} = 1.72$ V, $V_{min} = 0.125$ V):

$$\delta\Gamma_{open} = \frac{2\sqrt{2}\times 0.02\times 1.72\times 0.125}{(1.72+0.125)^2} \approx 0.004$$

For the 50 Ω termination ($V_{max} = 0.928$ V, $V_{min} = 0.900$ V):

$$\delta\Gamma_{50} = \frac{2\sqrt{2}\times 0.02\times 0.928\times 0.900}{(0.928+0.900)^2} \approx 0.014$$

This gives $\Gamma_{open} = 0.864 \pm 0.004$ and $\Gamma_{50} = 0.015 \pm 0.014$.

$$\delta Z_0 = \sqrt{\left(\frac{1-\Gamma_{50}}{1+\Gamma_{50}}\delta Z_L\right)^2 + \left(\frac{2Z_L}{(1+\Gamma_{50})^2}\delta\Gamma_{50}\right)^2} \tag{9}$$

Substituting $Z_L = 50\ \Omega$ and $\Gamma_{50} = 0.015$:

$$\delta Z_0 = \sqrt{\left(\frac{0.985}{1.015}\times 0.5\right)^2 + \left(\frac{100}{(1.015)^2}\times 0.014\right)^2} = \sqrt{(0.485)^2 + (1.359)^2} \approx 1.44\ \Omega$$

Thus, the calculated characteristic impedance is $Z_0 = 48.5 \pm 1.4\ \Omega$.

6.3 Inductance and capacitance per unit length

$$\frac{\delta L'}{L'} = \frac{\delta C'}{C'} = \sqrt{\left(\frac{\delta Z_0}{Z_0}\right)^2 + \left(\frac{\delta v_p}{v_p}\right)^2} \tag{10}$$

Substituting $Z_0 = 48.5\ \Omega$, $\delta Z_0 = 1.44\ \Omega$, and $v_p = 0.0122$ m/s:

$$\text{Fractional Error} = \sqrt{\left(\frac{1.44}{48.5}\right)^2 + (0.0122)^2} = \sqrt{(0.0297)^2 + (0.0122)^2} \approx 0.032 \text{ (or } 3.2\%)$$

Applying this fractional error to the calculated properties gives:

$$\delta L' = 0.032\times 2.44\times 10^{-7} \approx 0.08\times 10^{-7} \text{ H/m}$$
$$\delta C' = 0.032\times 1.04\times 10^{-10} \approx 0.03\times 10^{-10} \text{ F/m}$$

The final properties are $L' = (2.44 \pm 0.08)\times 10^{-7}$ H/m and $C' = (1.04 \pm 0.03)\times 10^{-10}$ F/m.

7 Results and Conclusion

- The phase velocity of the signal in the coaxial cable was determined to be $(2.15 \pm 0.03)\times 10^8$ m/s, giving a velocity factor of 0.715.
- The reflection coefficient for the open-ended cable was estimated at $\Gamma_{open} = 0.864 \pm 0.004$, whereas terminating the cable with a 50 Ω load reduced the coefficient to $\Gamma_{50} = 0.015 \pm 0.014$.

This significant reduction verifies that the cable is approximately matched to the 50 Ω termination and that its characteristic impedance is roughly $Z_0 = 48.5 \pm 1.4\ \Omega$.

- The inductance and capacitance per unit length were calculated as $L' = (2.44 \pm 0.08) \times 10^{-7}$ H/m and $C' = (1.04 \pm 0.03) \times 10^{-10}$ F/m, respectively.
- Finally, the attenuation per unit length was observed to be minimal but varied slightly across the 100 to 200 MHz frequency range (Table 1).

8 Questions to be answered

8.1 What is the significance of using 50 Ω termination in the experiment?

Terminating a transmission line with a load equal to its characteristic impedance prevents signal reflection. In this experiment, the 50 Ω termination was used to verify the cable's characteristic impedance; because the reflection coefficient dropped to nearly zero (0.015), it confirmed the cable is correctly matched to a 50 Ω load. The 50 Ω standard is a calculated engineering compromise between maximum power handling and minimum signal attenuation. Historical analysis of coaxial cable geometries demonstrated that a cable can carry maximum power before dielectric breakdown at an impedance of approximately 30 Ω, while it experiences the lowest possible signal loss at approximately 77 Ω. Since it is impossible to optimize for both extremes in a single cable, 50 Ω was selected as the ideal practical middle ground. This allows for robust power transmission while keeping signal loss low enough for efficient RF communication.

8.2 Why is the V_{rms} not same in consecutive minima?

The variation in V_{rms} at consecutive minima is primarily due to cable attenuation, which reduces the signal level as the wave propagates through the cable. The reflected wave loses energy traveling the length of the cable and back, meaning its amplitude is lower than the incident wave and it cannot perfectly cancel it out during destructive interference. Additionally, attenuation is frequency-dependent (as shown in Table 1), causing the amplitudes of the standing waves to decay at higher frequency nodes.

8.3 Why is a coaxial cable used?

A coaxial cable is used because its concentric design—a central conductor surrounded by an insulating dielectric and a metallic ground shield—confines the electromagnetic fields internally between the conductors. This unique structure prevents the internal signal from radiating outward and shields the transmission line from external electromagnetic interference, making it highly efficient for transmitting high-frequency RF signals with minimal loss.

9 Key Takeaways

Studying a transmission line like the coaxial cable was helpful in understanding the requirements for a material to be a good choice for transmission. Also, I was able to determine the velocity of EM

waves inside a transmission line.